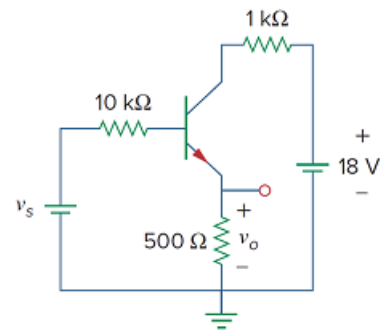


Problem

Calculate v_s for the transistor in Fig. 3.126 given that $v_o = 6\text{ V}$, $\beta = 90$, $V_{BE} = 0.7\text{ V}$.

Figure 3.126:



Step-by-step solution

Step 1 of 4

Refer to the circuit in Figure 3.126 in the textbook.

Consider the circuit diagram shown in Figure 1.

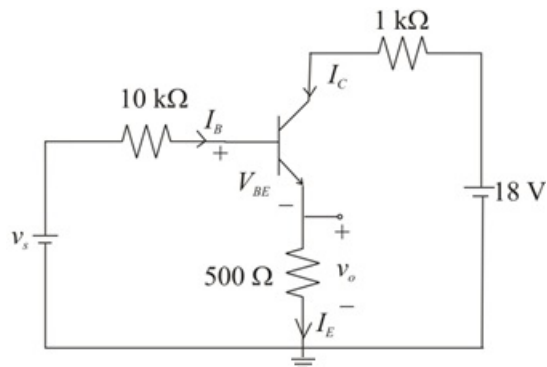


Figure 1

Step 2 of 4

Calculate the value of the emitter current I_E .

$$I_E = \frac{v_o}{500}$$

Substitute 6 V for v_o .

$$I_E = \frac{6}{500} \\ = 12\text{ mA}$$

Therefore, the value of the current I_E is 12 mA .

Step 3 of 4

Calculate the value of the base current I_B .

$$I_B = \frac{I_E}{1 + \beta}$$

Substitute 12 mA for I_E and 90 for β .

$$\begin{aligned} I_B &= \frac{12 \text{ mA}}{1 + 90} \\ &= \frac{12 \text{ mA}}{91} \\ &= 132 \text{ } \mu\text{A} \end{aligned}$$

Therefore, the value of the base current I_B is 132 μA .

Step 4 of 4

Apply Kirchhoff's Voltage Law for the input loop.

$$-v_s + I_B (10 \text{ k}) + V_{BE} + v_o = 0$$

Substitute 132 μA for I_B , 6 V for v_o and 0.7 V for V_{BE} .

$$-v_s + (132 \text{ } \mu\text{A})(10 \text{ k}) + 0.7 + 6 = 0$$

$$\begin{aligned} v_s &= 1.32 + 6.7 \\ &= 8.02 \text{ V} \end{aligned}$$

Therefore, the value of the voltage v_s is, 8.02 V.